

Adarsh Agarwala

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building AI-powered applications and developer tools, including LLM-powered RAG pipelines with hallucination mitigation, self-verifying retrieval systems, and full-stack products shipped end to end. Skilled at turning state-of-the-art AI/ML models (LLMs, NLP) into deployed, workflow-automating systems, and comfortable collaborating closely with product, engineering, and leadership to ship measurable efficiencies. Strong mathematical foundation in linear algebra, probability, and statistics, complemented by competitive programming expertise in C++.

EDUCATION

Indian Institute of Information Technology (IIIT), Pune

Pune, India

Bachelor of Technology in Computer Science and Engineering

Expected May 2029

- CGPA: 9.38/10.0 SGPA: 9.6/10.0
- Relevant Coursework: Linear Algebra, Differential Calculus, Probability & Statistics, Deep Learning, Data Structures & Algorithms.

TECHNICAL SKILLS

Languages: Python (Advanced), C++ (Advanced), JavaScript (ES6+), TypeScript

Full-Stack & Web: React.js, Next.js, Node.js, Express.js, FastAPI, MongoDB, Tailwind CSS, REST APIs, WebSockets

Deep Learning & ML Frameworks: PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, Transformers (HuggingFace), sentence-transformers, OpenCV

AI & LLM Techniques: Retrieval-Augmented Generation (RAG), Hybrid & Vectorless RAG (BM25 + Dense Embeddings), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, Tool-Calling Agents, Conversational Memory, NLI-based Hallucination Detection, LoRA Fine-Tuning

Developer Tooling & Desktop: Electron, Chrome Extensions (Manifest V3), node-pty, xterm.js, IPC / contextBridge, MutationObserver, SQLite

System Design: RESTful API design, microservices architecture, caching strategies (Redis), message queues, load balancing, SQL/NoSQL database design, CAP theorem

Tools & Platforms: Git, Linux, OpenAI API, Groq, LangChain, LangGraph, ChromaDB, Ollama, Docker (basic), Vercel

PROJECTS

Medical RAG: Trust-Aware Q&A | *Python, FastAPI, ChromaDB, Transformers (NLI), Ollama*

[GitHub](#)

- Built a **local-first retrieval-augmented generation system** that answers medical questions **only from a document corpus**, running fully on-device with local embeddings and a local LLM via **Ollama** so no data leaves the machine.
- Implemented **hybrid retrieval** combining BM25 sparse search with dense embeddings, fused via **Reciprocal Rank Fusion (RRF)** and refined with a **cross-encoder reranker** to surface the most relevant source passages.
- Added a **claim-level NLI verification layer** that splits each answer into atomic claims and labels every claim SUPPORTED / WEAK / UNSUPPORTED / CONTRADICTED, producing a **trust score** that flags likely hallucinations.
- Built an **evaluation harness** (recall@k, MRR, nDCG@k, hallucination rate, groundedness) with hybrid-vs-dense and reranker ablations; debugged a metric bug (nDCG > 1.0 from duplicate chunk sources) and added regression tests to lock the fix.
- Shipped a **streaming FastAPI (SSE)** backend and web UI showing the answer, trust badge, per-claim verification, and citations, with a safety guardrail that refuses personal medical-advice requests.

amber Copilot: AI Student Housing Assistant | *Python, FastAPI, Groq/Llama 3.3, Redis, Docker*

[GitHub](#)

- Identified that **amberstudent.com**'s existing accommodation copilot relied on a rigid, manual, form-based flow with poor UX, and built an **LLM-driven conversational booking assistant** to replace it, letting international students search, compare, and book housing across all supported cities in natural language.
- Designed a **tool-calling agent loop** (FastAPI) that grounds every reply in live data from 5 amber API endpoints with citation tracking and grounding guardrails, eliminating hallucinated properties or prices.

- Built a **4-layer conversational memory** system (turn buffer, rolling summary, structured `SearchState` slots, property shortlist) with token budgeting, plus deterministic NLU slot extraction and reference resolution (e.g. “compare the top 2”) for coherent multi-turn booking.
- Streamed responses over **SSE** into an embeddable React/TypeScript widget (Shadow DOM) with structured property cards; powered by **Groq** inference (Llama 3.3 70B agent + 8B router) behind a swappable model interface with provider failover, and containerized via **Docker Compose** (FastAPI, React, Redis).

AgentWatch: Agent Observability | *Electron, TypeScript, node-pty, React, SQLite, MV3*

[GitHub](#)

- Architected a cross-platform **Electron desktop app** that wraps long-running AI CLI agents (Gemini CLI, Claude Code) in a **non-intrusive node-pty interception layer**, mirroring the terminal in real time with live state annotations (thinking / reading / writing / waiting / done) without modifying agent behavior.
- Built a **permission-gating control plane** that detects blocking prompts (file writes, network calls) in the output stream, surfaces Allow/Deny UI, and injects responses into stdin to unblock the agent in one click, backed by a **SQLite audit log** (sessions, events, verdicts) for replay.
- Engineered a **secure IPC bridge** (Electron contextIsolation + preload) exposing a typed `window.agentwatch` API via `contextBridge` between the main-process PTY manager and the React renderer.
- Shipped a companion **Manifest V3 Chrome extension** monitoring 20+ browser AI agents (Claude, ChatGPT, Gemini, Perplexity, Cursor, and more) with a **layered completion-detection engine** (per-site button-state adapters + MutationObserver quiescence fallback) and one-click cross-window jump-to-tab, using least-privilege, domain-scoped permissions.

Portfolio v2 & AI Assistant | *Next.js 14, TypeScript, Tailwind, Framer Motion*

[Live](#) | [GitHub](#)

- Designed and shipped a production portfolio on the **Next.js 14 App Router** with server routes and dynamic project/experience pages, motion-driven UI via Framer Motion, and a built-in admin CMS for live content editing.
- Engineered an in-browser **AI assistant** grounded on a personal knowledge base using a **hybrid, vectorless RAG** pipeline (TF-IDF retrieval) feeding an LLM through a serverless API route, with a guardrail system prompt that restricts answers to on-topic, source-grounded responses.
- Added voice input/output via the Web Speech API, a live answer-source indicator, and end-to-end type safety; deployed on **Vercel** with CI/CD from GitHub.

ACHIEVEMENTS & LEADERSHIP

Competitive Programming Champion: Secured 1st Place in the First-Year Competitive Programming Competition at IIIT Pune; active Codeforces competitor (Handle: AdarshAg), solving advanced graph, dynamic programming, and data structure problems in C++.

Smart India Hackathon (SIH): Mentored a competitive team through ideation and full development phases of a national-level hackathon, coordinating cross-functional collaboration under strict time and resource constraints.

Hackathons & Innovation: Participated in the Indian Innovators Challenge; led Team SHA256 in architecting an optimized backend infrastructure during a high-pressure 48-hour competitive engineering sprint.